Industrial Internship Report on

"AgriVision: AI-Powered Crop and Weed Detection"

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was AgriVision, an AI-powered crop and weed detection system. It uses a fine-tuned YOLO11n object detection model and a Streamlit web dashboard to find crops and weeds in field images, show bounding boxes with confidence scores, and give general crop-protection guidance. On the validation set the model reached 85.1% mAP@50, with 81.9% precision and 78.7% recall.

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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Preface

Technology for Divyangjan, Kanpur, I spent the six weeks of this internship designing, training, testing and deploying AgriVision, a computer-vision application that detects crops and weeds in field images. The work covered the full machine-learning life cycle: understanding the problem, studying the data, fine-tuning a YOLO11n object detection model, building a Streamlit dashboard around it, evaluating the results and publishing the app online.

A relevant internship is important for career development because classroom learning rarely shows how much work lies between a working model and a usable product. Working on a real problem taught me to deal with practical limits such as image quality, model errors, deployment constraints and clear communication of results, which are the skills industry expects from a fresh engineer.

My project addresses weed detection in agriculture. Weeds compete with crops for water, light and nutrients, and finding them by walking through a field is slow and subjective. AgriVision lets a user upload a photo of a field, runs a trained object detector, draws a box around every crop plant and weed it finds, reports confidence scores and counts, and offers general crop-protection guidance.

This opportunity was provided by upskill Campus and The IoT Academy in association with UniConverge Technologies Pvt Ltd (UCT). It gave me a structured way to work on an industrial-style problem statement with a fixed timeline, a working deliverable and a final report.

The program was planned as a six-week schedule, shown in the figure below: understanding the problem statement, studying the technology, building and training the solution, testing it, and finally preparing the project submission and this report.



Through this project I learned how to fine-tune and evaluate an object detection model, read metrics such as precision, recall and mAP, and turn a trained model into a web application. The experience was very rewarding and improved my confidence in solving open-ended problems.

I would like to thank upskill Campus, The IoT Academy and UniConverge Technologies Pvt Ltd for this opportunity, and my mentors, faculty members, friends and family who guided and supported me during the internship. [Add the names of the people you wish to thank here.]

To my juniors and peers: start early, pick a problem you are curious about, build it end to end, and do not be discouraged by poor early results; every iteration teaches something.

2 Introduction

2.1 AboutUniConverge Technologies Pvt Ltd

Acompanyestablishedin 2013 andworking in DigitalTransformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

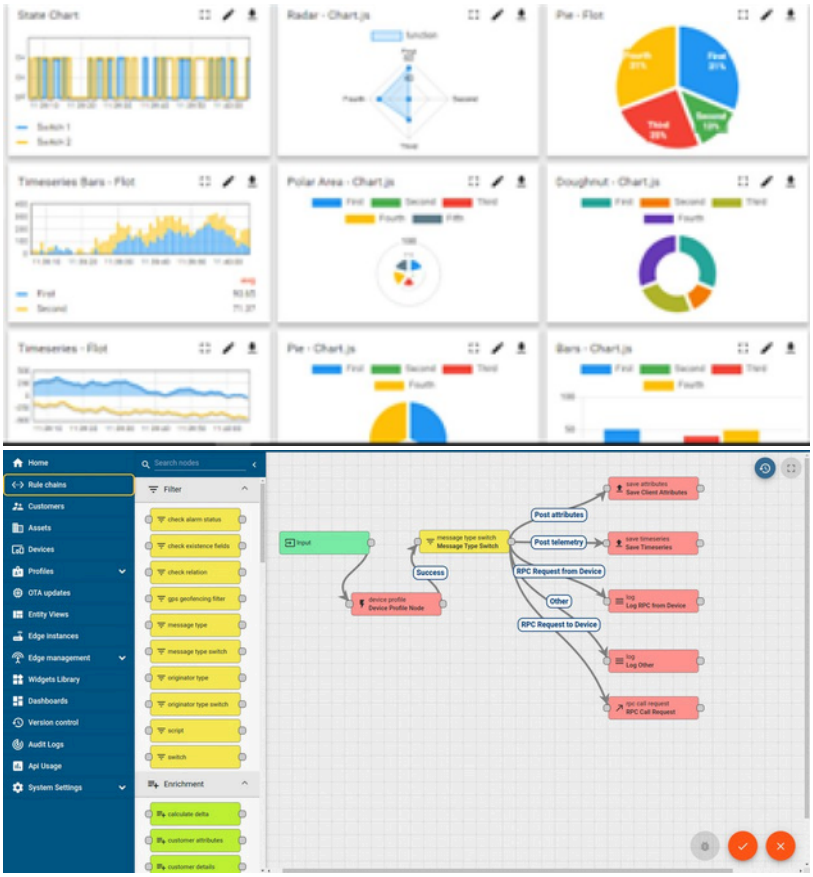


i. UCT IoT Platform (uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to
- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



ii. Smart Factory Platform (FACTORY WATCH)

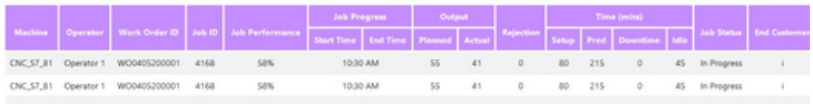
Factorywatchisa platform forsmartfactoryneeds.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.







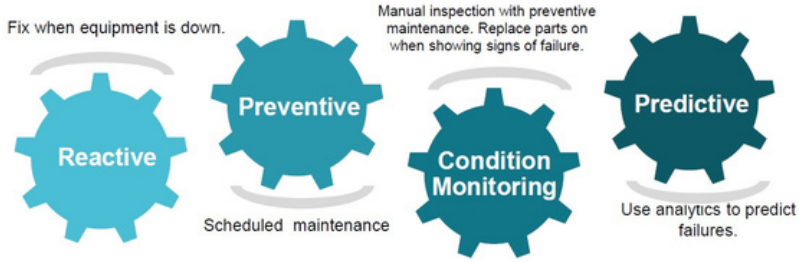
LoRaWAN™

iii. based Solution

UCT is one of the early adopters of LoRAWAN teschnologyand providing solution in Agritech, Smart cities, IndustrialMonitoring,Smart StreetLight, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

UCT is providing Industrial Machinehealthmonitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

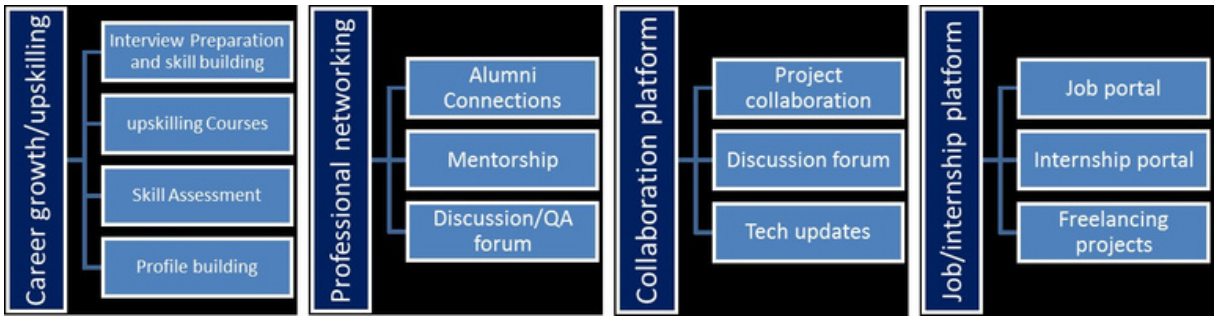
upskill Campus alongwith The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

https://www.upskillcampus.com/

upSkill Campus aiming to upskill 1 million learners in next 5 year



2.3 The IoT Academy

TheIoTacademy is EdTechDivision of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objectivefor thisinternshipprogram was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

[1] Ultralytics,"YOLO11 documentation." https://docs.ultralytics.com/models/yolo11/

[2] Streamlit Inc., "Streamlit documentation." https://docs.streamlit.io

[3] J. Redmon, S. Divvala, R. Girshick and A. Farhadi, "You Only Look Once: Unified, Real-Time Object Detection," IEEE CVPR, 2016.

[4] PyTorch, "PyTorch documentation." https://pytorch.org/docs

[5] K. Singh, "AgriVision project source code." https://github.com/Khushi-Singh7/upskillcampus

2.6 Glossary

Terms	Acronym / Meaning
AI	Artificial Intelligence
ML	Machine Learning
YOLO	You Only Look Once (real-time object detection model family)
YOLO11n	Nano (smallest) variant of the YOLO11 model by Ultralytics
IoU	Intersection over Union (overlap between two bounding boxes) Non-
NMS	Maximum Suppression (removes duplicate overlapping detections) mean
mAP	Average Precision (standard object detection accuracy metric) upskill
USC	Campus
UCT	UniConverge Technologies Pvt Ltd

3 Problem Statement

In the assigned problem statement, the task was to build a practical AI solution for agriculture that can lookat an image of a field and automatically tell crop plants and weeds apart.

Weeds compete with crops for sunlight, water and nutrients and, if they are not controlled in time, reduce yield and quality. On most farms weed monitoring is still done by walking through the field and judging by eye, which is slow, tiring, inconsistent between people and hard to do over large areas. Spraying the whole field with herbicide is a common response, but it wastes chemicals, raises cost and harms the environment.

The problem I worked on was therefore: given a photograph of a field, automatically locate each crop plant and weed with a bounding box and a confidence score, summarise the result in a form a farmer or agronomist can understand, and show general crop-protection guidance, all through a simple web interface that needs no technical knowledge. The solution had to:

- detect two classes, crop and weed, in an uploaded image;
- let the user adjust the confidence threshold, the NMS IoU threshold and the inference image size;
- run in an ordinary web browser, without installing anything; and
- be honest about its limits, so that it is not mistaken for a diagnosis or a pesticide prescription.

Existing and Proposed solution

- **Manual scouting:** workers walk the field and inspect plants by eye. It needs no technology but is slow, labour intensive and depends on the experience of the person.
- **Uniform herbicide spraying:** the whole field is treated whether or not weeds are present, which wastes chemicals and increases cost and environmental impact.
- **Commercial precision-agriculture systems:** sensor-equipped sprayers and drone or satellite analytics can be accurate, but they usually need special hardware, training and budget, which puts them out of reach of small farms and students.
- **General image-classification or plant-identification apps:** they label a whole picture or a single leaf, so they do not show where the weeds are in a field or how many there are.

Proposed solution. AgriVision is a web application built around a fine-tuned YOLO11n object detection model (Ultralytics, PyTorch). The user selects the crop, uploads a photo and, if needed, adjusts the confidence threshold, NMS IoU threshold and inference image size. The app returns the annotated image with boxes and confidence scores, the number of crops and weeds, the weed share of all detections, a table of detections with bounding-box coordinates, general crop-protection guidance and an exportable report. It is deployed on Streamlit Community Cloud, so anyone can use it from a browser.

Value addition.

- **Detection instead of classification:** every crop plant and weed is located with its own box, so the user can see where the problem is and count it.
- **Lightweight and low cost:** YOLO11n is the smallest YOLO11 variant, which keeps the app fast enough for ordinary CPU hosting; no special hardware is required.
- **Transparent results:** confidence scores, adjustable thresholds and a note that counts describe only the latest analysis, not every plant in the field.
- **Responsible guidance:** the crop-protection tab gives general preventive and weed-management practices, and clearly says it is not a diagnosis or pesticide prescription.
- **Easy to use:** a three-step flow (upload, detect, act) with light and dark themes.

4.1 Code submission (Github link)

https://github.com/Khushi-Singh7/upskillcampus

Live application: https://agrivision-crop-weed-detection-eycrbf3di56s6sqrc3cuxg.streamlit.app/

4.2 Report submission (Github link)

https://github.com/Khushi-Singh7/upskillcampus/blob/main/reports/AgriVision_Industrial_Internship_Report.pdf

Proposed Design/ Model

(confidence filter and non-maximum suppression), and finally visualisation and guidance. The code is split into a presentation layer (app.py, built with Streamlit), an inference layer (backend/predict.py), a crop-protection knowledge layer (backend/recommendations.py) and the trained weights (models/best.pt).

5.1 High Level Diagram (if applicable)

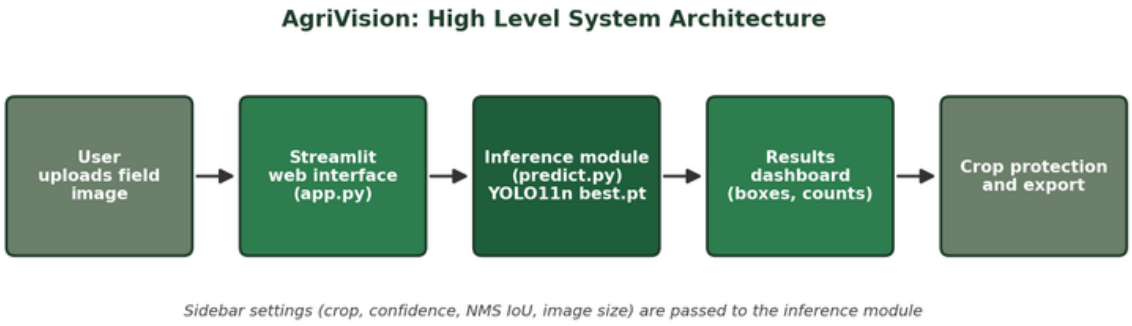


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

The user interacts only with the Streamlit interface. The interface passes the uploaded image and the sidebar settings to the inference module, which loads the trained YOLO11n weights and returns the detections. The results are shown on the dashboard, and the selected crop is used to show the matching crop-protection guidance and export the report.

5.2 Low Level Diagram (if applicable)

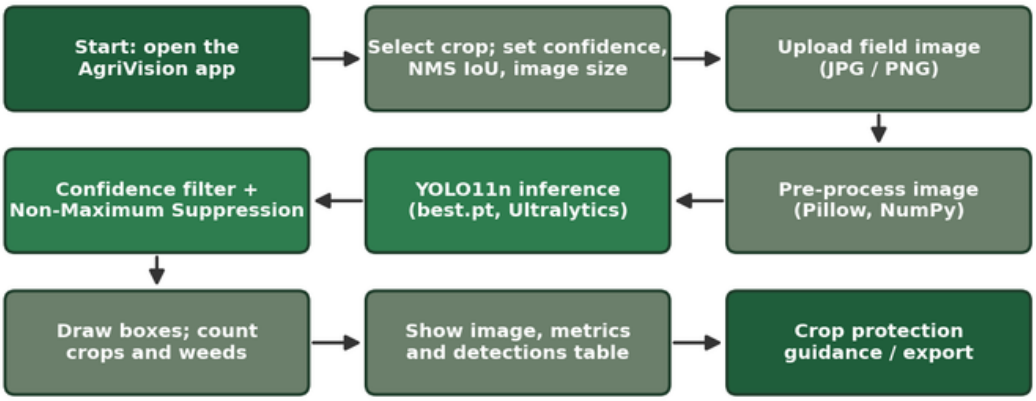


Figure 2: LOW LEVEL FLOW OF AN ANALYSIS

Each analysis is one pass through this flow. The confidence threshold removes weak predictions, and non-maximum suppression removes duplicate boxes that overlap by more than the chosen IoU value. The counts shown in the dashboard come only from the boxes that survive both steps.

5.3 Interfaces(if applicable)

The application is a single web page with four sections (Overview, Detect & Analyze, Crop Protection and About Project) and a sidebar for analysis settings. The browser talks to the Streamlit server over HTTPS; the image is processed on the server and only the results are sent back. The sidebar contains:

- **Crop being grown:** selects the crop (Wheat by default) used for the guidance text.
- **Confidence threshold:** minimum score for a detection to be kept (default 0.25).
- **NMS IoU threshold:** overlap above which duplicate boxes are merged (default 0.50).
- **Inference image size:** size to which the image is resized for the model (default 640).

The screenshots below show the interface.

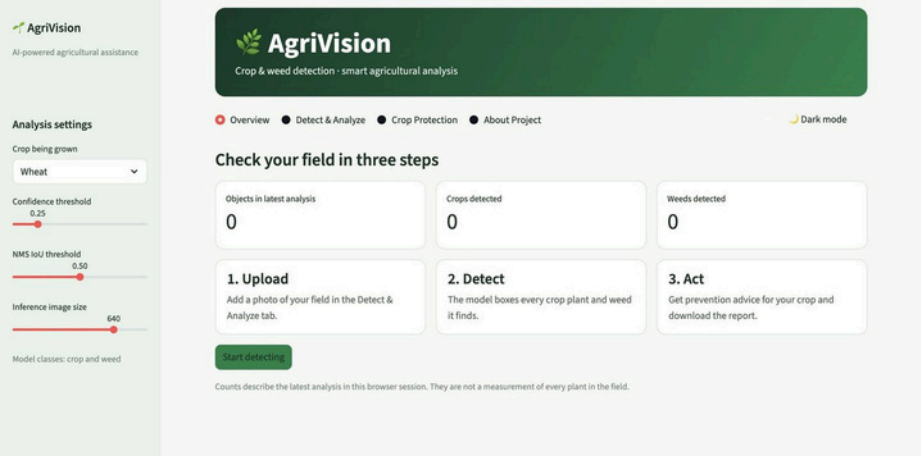


Figure 3: OVERVIEW PAGE (LIGHT MODE) WITH SIDEBAR SETTINGS

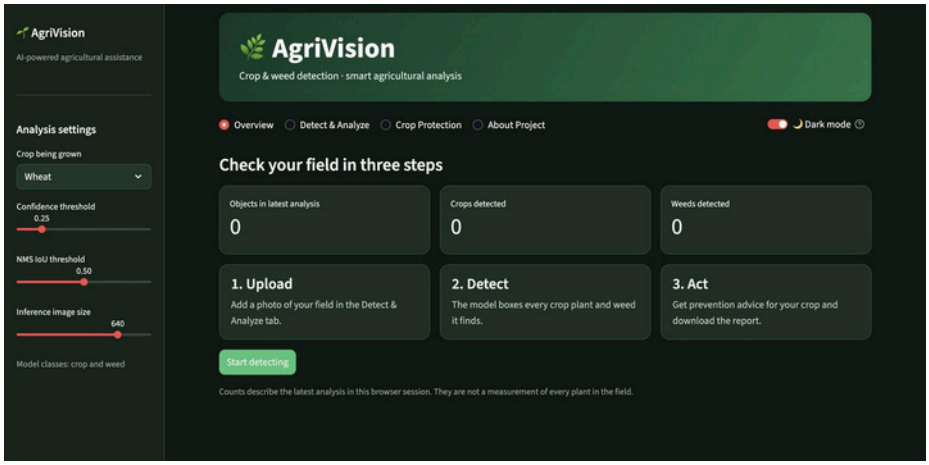


Figure 4: OVERVIEW PAGE IN DARK MODE

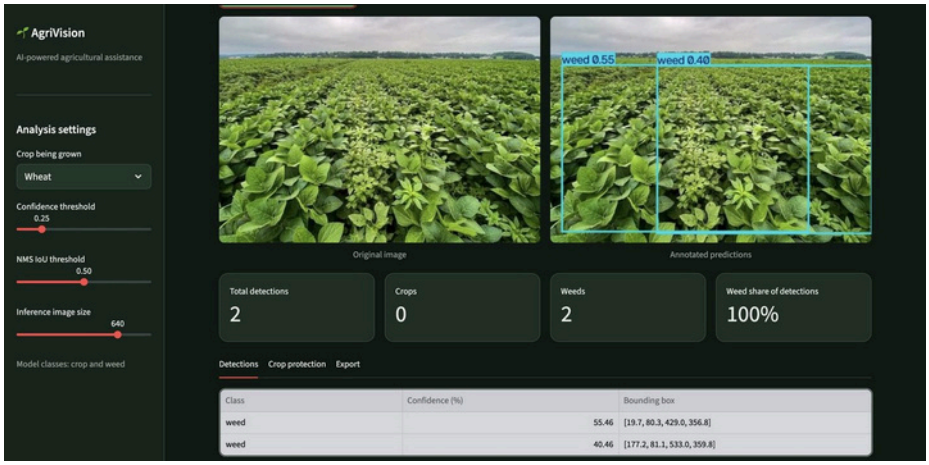


Figure 5: DETECT AND ANALYZE: ORIGINAL AND ANNOTATED IMAGE, COUNTS AND DETECTIONS TABLE

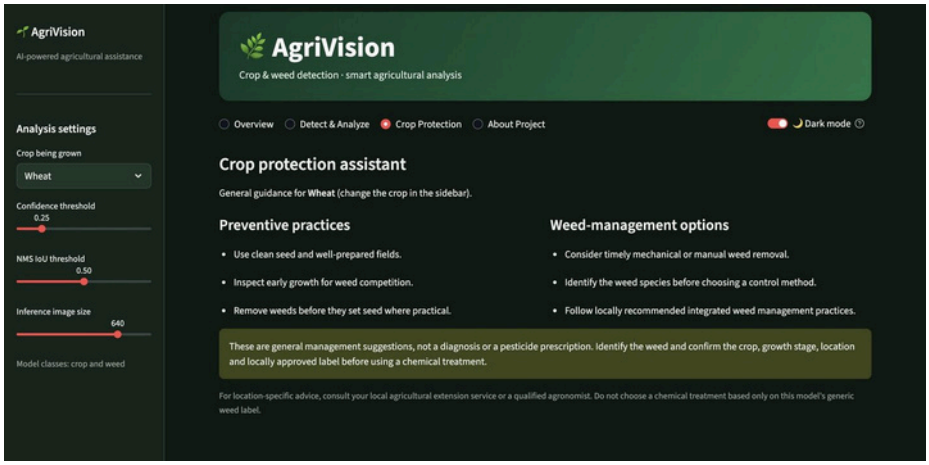


Figure 6: CROP PROTECTION ASSISTANT

6 Performance Test

AgriVision is meant foruse on real field images, so its performance was judged against practical constraints and not onlyas an academic exercise. The constraints identified were:

- **Detection accuracy:** the model must find weeds and crops reliably (precision, recall, mAP).
- **Speed and model size:** the app runs on a shared, CPU-only cloud server, so it needs a small model and a moderate input size.
- **Image quality:** lighting, camera angle, plant density and growth stage change the result.
- **Threshold sensitivity:** the confidence and NMS thresholds trade missed detections against false alarms.
- **Scope of the model:** only two generic classes are detected, so species, diseases and pesticide choice are out of scope.

How these constraints were handled in the design.

- The smallest variant, YOLO11n, was chosen so that inference stays light, and the default inference image size is 640.
- The confidence threshold, NMS IoU threshold and image size are exposed as sliders, so the user can tune the model to the image.
- The interface states that counts describe only the latest analysis, and the crop-protection page states that its advice is general and not a diagnosis or pesticide prescription.

Test results. The accuracy results on the validation set and the observations from a sample field image are given in section 6.3. Response time per image was not formally benchmarked during the internship; this should be measured on the target hardware before wider use.

6.1 Test Plan/ Test Cases

No.	Test case	Expected result
T1	Evaluate the trained model on the validation set (260 images, 380 labelled instances)	Precision, recall, mAP@50 and mAP@50-95 are computed for the crop and weed classes
T2	Upload a field image with default settings (confidence 0.25, NMS IoU 0.50, size 640)	Annotated image with boxes and scores, counts and a detections table are shown
T3	Change the confidence, NMS IoU and image size sliders	The sliders are available and the analysis uses the chosen values
T4	Open the app in light and dark mode	All sections remain readable in both themes
T5	Open the Crop Protection page for the selected crop	Preventive practices, weed-management options and the disclaimer are shown

6.2 Test Procedure

Thetestswerecarried out in the following order:

- The YOLO11n model was fine-tuned on labelled crop and weed images and evaluated on the held-out validation set to obtain precision, recall, mAP@50 and mAP@50-95 (T1).
- The application was deployed on Streamlit Community Cloud and opened through its public link.
- A sample field image was uploaded with the default sidebar settings, and the annotated image, the metric cards and the detections table were inspected (T2).
- The sidebar controls were checked, the dark-mode toggle was used, and the Crop Protection page was opened (T3 to T5).

6.3 Performance Outcome

Modelaccuracy (T1). Thetrainedmodel gave the following results on the validation set of 260 images with 380 labelled instances:

Metric	Result
Precision	81.9%
Recall	78.7%
mAP@50	85.1%
mAP@50-95	59.3%

Precision of 81.9% means that about 82 of every 100 drawn boxes are correct, and recall of 78.7% means that about 79 of every 100 labelled objects are found. The mAP@50 of 85.1% shows good detection at a loose overlap requirement, while the lower mAP@50-95 of 59.3% shows that boxes are less tight under a stricter one. These values apply to the validation data; real-field results will vary with image quality, lighting, camera angle and field conditions.

Sample analysis (T2). For the field image in Figure 5 the app returned two detections, both of class weed, and no crop detections, so the weed share of detections was 100%:

Class	Confidence(%)	Boundingbox[x1, y1,x2,y2]
weed	55.46	[19.7, 80.3, 429.0, 356.8]
weed	40.46	[177.2, 81.1, 533.0, 359.8]

This image had no ground-truth labels, so this is a qualitative observation, not a measured error. Both boxes are large and overlap heavily over a dense leafy canopy, and no crop was reported although the image appears to show mostly crop plants. This suggests that dense, close-up or oblique images are harder for the model, and that a result should be checked against the image before acting on it.

Interface and guidance (T3 to T5). The sliders, both themes (Figures 3 and 4) and the Crop Protection page (Figure 6) worked as expected; slider sweeps were not recorded.

7 My learnings

Thisinternship gave mehands-on experience of taking a machine-learning idea all the way to a working, public application. My main learnings were:

- **Object detection:** how YOLO-style detectors work, and how confidence scores, IoU and non-maximum suppression shape the final boxes.
- **Evaluation:** how to read precision, recall, mAP@50 and mAP@50-95, and why a good metric on a validation set does not guarantee good results on every real image.
- **Application building and deployment:** building an interactive dashboard with Streamlit, separating the interface from the inference code, managing dependencies with requirements.txt and packages.txt, and deploying on Streamlit Community Cloud.
- **Version control:** organising a project in a GitHub repository with a clear structure and README.
- **Responsible AI:** being clear about what the model cannot do, for example that it does not identify species or diseases and must not be used to choose a pesticide.
- **Professional skills:** planning work against a deadline, debugging, and writing a clear report.

These skills will help me in my career in data science and AI, where building a model is only part of delivering a useful and trustworthy product.

8 Future work scope

Ideas that could not be taken up because of the time limit, but can be done in future:

- Train on a larger and more varied dataset (more crops, growth stages, lighting, camera angles and dense canopies) to reduce the missed crops and oversized boxes seen in the sample analysis.
- Add crop-specific and weed-species classes so that the advice can be matched to the actual weed.
- Process video and drone footage and map the weed locations on a field map.
- Measure and optimise inference time, for example with model export to ONNX or quantisation, and add a mobile-friendly or offline version.
- Estimate weed coverage (area) instead of only counting boxes, to support spot spraying.
- Add multi-language support and region-specific guidance in cooperation with local agricultural extension services.
- Add automated tests and a record of analyses over time so that field progress can be tracked.